

Why a track graph

Detailed routers do not hop GCell to GCell

The idea

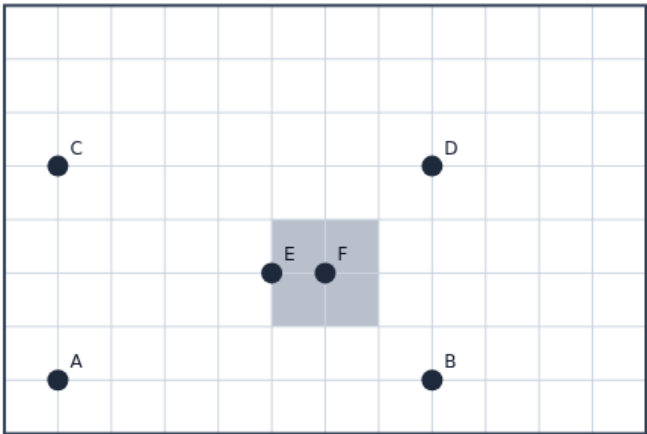
- A directed M2 edge upward when $y + 1$ is less than n_y
- Canonical keys sort endpoints so left-to-right and bottom-to-top always win
- On twelve by eight you get eighty-eight M1 edges and eighty-four M2 edges

12x8 grid

ROUTING GRID

12x8 grid

Step 1 / 5 · grid · module01-01-routing-grid



Explanation

Detailed routes walk grid points; M1 horizontal, M2 vertical.

- Cap=2 per track
- Blockage 2x2

12x8

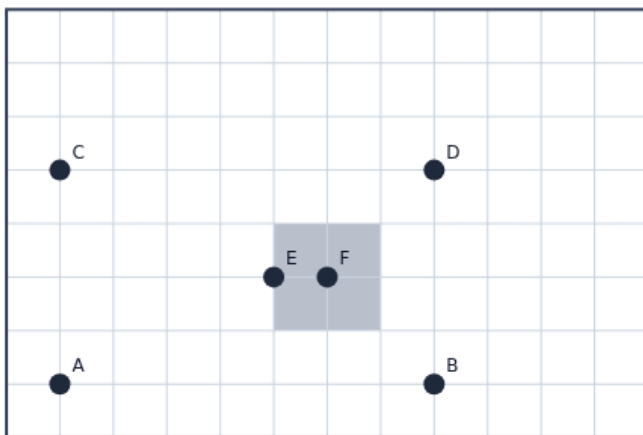
12x8 grid

Two layers

ROUTING GRID

Two layers

Step 2 / 5 · layers · module01-01-routing-grid



Explanation

Horizontal moves on M1; vertical on M2; via switches at grid point.

- M1 / M2
- Layer rules

Not GCell

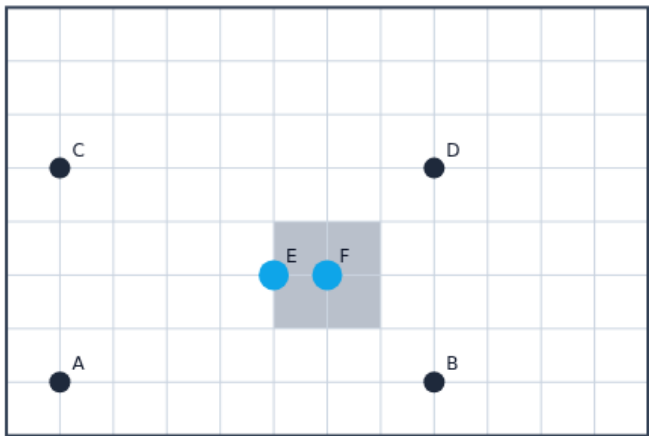
Two layers

Blockage

ROUTING GRID

Blockage

Step 3 / 5 · block · module01-01-routing-grid



Explanation

Cells (5-6, 2-3) are blocked—maze must detour.

- 2x2 macro
- Pin nudge

Lee lab

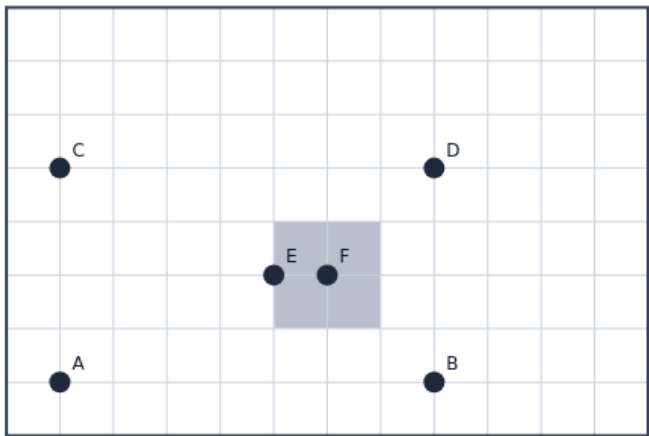
Blockage

Neighbors

ROUTING GRID

Neighbors

Step 4 / 5 · neigh · module01-01-routing-grid



Explanation

Free grid points have up to four neighbors.

- Four-connected

mid=4

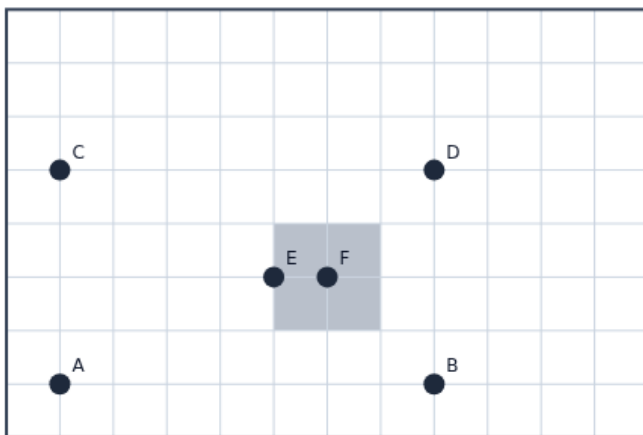
Neighbors

Capacity

ROUTING GRID

Capacity

Step 5 / 5 · cap · module01-01-routing-grid



Explanation

Track usage vs capacity drives overflow and rip-up.

- Cap=2

Document goldens

Capacity

Browser lab track

- Open the **routing-grid** lab
- Toggle layer M1 and M2 overlays
- Highlight one horizontal track between columns one and two on row one
- Read the track list in the metrics panel and match the counts

Implement track

- Inspect ``h_edge``, ``v_edge``, and ``track_key`` in ``common/drutil.py``
- Print sample M1 and M2 keys for `tiny_dr`
- Verify diagonals are absent and layers match movement axis

Pitfalls

- Counting grid points instead of directed tracks
- Treating M1 and M2 as interchangeable
- Forgetting blockages still occupy cells even if tracks exist between free neighbors

Your turn

- Finish the checklist
- Sketch M1 on one row from memory
- Next: map pin placements to access points on the grid